

APPENDIX G: EarthBank Project

Summary



National Geochemical, Isotopic, Geochronological &
Thermochronological Data Platform

GDAC-SA Advanced Analytics Platform RFQ | Lithodat Pty Ltd

Project Overview: EarthBank (formerly AusGeochem) is Australia's national cloud-based FAIR-compliant geoscience data repository, developed and operated by Lithodat Pty Ltd in partnership with AuScope Geochemistry Network. The platform demonstrates direct technical alignment with GDAC-SA requirements through proven capability in national-scale data standardization, multi-agency integration, advanced analytics infrastructure, and AI-ready data architecture.

1. Executive Summary

1.1 What is EarthBank?

EarthBank is an Australian national cloud-based public data repository for geochemistry, geochronology, isotopes, thermochronology, and geological sample metadata. Lithodat designed, built, and operates the entire platform under contract to AuScope Geochemistry Network (AGN), funded through Australia's National Collaborative Research Infrastructure Strategy (NCRIS).

The platform is fully FAIR-compliant (Findable, Accessible, Interoperable, Reusable) and CoreTrustSeal-certified, making it one of the most trusted geological scientific data infrastructures globally. EarthBank serves as Australia's centralized repository for analytical geoscience data, unifying datasets from federal and state geological surveys, universities, and research institutions.

AuScope NCRIS Funding Context

EarthBank is part of AuScope, Australia's national geoscience research infrastructure organization established in 2006. AuScope has received **over \$100 million in cumulative NCRIS funding** since inception, including:

- **\$42.8 million** initial NCRIS allocation (2006) for integrated national geoscience infrastructure
- **\$45 million** NCRIS RIIP funding (2023) for Earth, coastal, Antarctic monitoring, drilling, data, and modelling
- **\$19 million** specifically allocated for EarthBank expansion, including 10 EarthBank fellows

This substantial government investment demonstrates Australia's recognition of national geoscience data infrastructure as critical research infrastructure—exactly the paradigm GDAC-SA seeks to implement for Saudi Arabia.

1.2 Relevance to GDAC-SA Project

EarthBank demonstrates Lithodat's proven capability to deliver the exact type of national geoscience data infrastructure required by the Saudi Geological Survey for the GDAC Advanced Analytics Platform. The project showcases:

- **National-Scale Data Integration:** Successfully integrated >12 million analytical results from multiple federal and state geological surveys, demonstrating capability to harmonize Saudi Arabia's distributed geological datasets
- **Multi-Agency Coordination:** Proven ability to coordinate data standards and workflows across 10+ Australian universities and multiple government agencies, directly applicable to GDAC's multi-stakeholder environment
- **Unified Data Models:** Custom-built relational database architecture with unified data models across geochemistry, geochronology, isotopes, and thermochronology—the same types of analytical data GDAC-SA needs to integrate

- **Advanced Analytics Infrastructure:** REST API services, machine learning-ready data structures, and cloud-native architecture optimized for the advanced analytics capabilities required by GDAC Phase 1
- **FAIR & Quality Standards:** CoreTrustSeal certification and ISO security alignment demonstrate the data governance rigor required for national geological survey infrastructure
- **Scalability Proven:** System handles 350,000+ registered samples, 1,300+ active users, and continues to scale as national datasets grow

Key Alignment with GDAC-SA Phase 1: EarthBank's architecture directly addresses GDAC's core requirements: (1) National data standardization and cleansing, (2) Multi-source data integration and harmonization, (3) Advanced analytics and ML-ready infrastructure, (4) Secure cloud-based deployment, (5) Multi-agency collaboration frameworks, and (6) Internationally recognized data quality standards.

2. AI and Machine Learning Capabilities

2.1 AI-Ready Data Architecture

EarthBank was designed from inception to support advanced analytics and machine learning applications. The platform's data architecture includes:

- **Standardized Data Schemas:** Custom SKOS-format relational database architecture ensures consistent, machine-readable data structures optimized for ML algorithms
- **REST API Access:** Full programmatic access to all datasets enables integration with external ML workflows, Python-based analytics tools, and automated data pipelines
- **Spatial-Temporal Analytics:** 4D data structures (x, y, z, time) with paleogeographic reconstruction capabilities support temporal machine learning models for mineral prospectivity
- **Quality Metadata:** Automated QA/QC pipelines and comprehensive procedural metadata enable ML models to assess data quality and uncertainty
- **Cloud-Native Infrastructure:** 100% AWS-hosted architecture provides scalable compute resources for on-demand ML processing

2.2 Published Advanced Analytics Applications

Lithodat's team has published peer-reviewed research demonstrating advanced analytics applications using EarthBank data:

Machine Learning for Mineral Prospectivity

Publication: Farahbakhsh et al. (2025). Machine learning-based spatio-temporal prospectivity modeling of porphyry systems. Tectonics.

Application of machine learning algorithms to 4D geochemical and geochronological datasets from EarthBank, demonstrating the platform's capability to support advanced ML-based mineral exploration targeting. This directly addresses GDAC-SA's Phase 1 advanced analytics requirements.

Big Data Thermochronology Analytics

Publication: Boone et al. (2023). A geospatial platform for tectonic interpretation of low-temperature thermochronology Big Data. Scientific Reports, 12.

Lithodat team (Kohlmann, Noble, Theile) co-authored this publication demonstrating a geospatial platform for processing and analyzing large-scale thermochronology datasets. The platform automatically processes analytical results and generates tectonic interpretations—showcasing the type of automated analytics GDAC-SA Phase 1 envisions.

Automated Analytical Workflows

Publication: Peternell et al. (2009). A new approach to crystallographic orientation measurement for apatite fission track analysis: effects of crystal morphology and implications for automation. Chemical Geology, 265,

Fabian Kohlmann co-authored research on automation of apatite fission track analysis, demonstrating technical innovation in analytical geochemistry workflows.

2.3 ML Infrastructure Features

ML Capability	EarthBank Implementation	GDAC-SA Applicability
Data Pipelines	Automated ingestion, QA/QC, and transformation	Saudi datasets automated processing
API-First Design	REST API for Python, R, Julia integration	Integration with GDAC analytics tools
Spatial Analytics	IDW interpolation, swath profiles, 3D visualization	Mineral prospectivity mapping for Saudi Arabia
Temporal Analytics	LithoPlates: 11 plate models to 1.8 Ga reconstruction	Deep-time resource prediction modeling
Cloud Compute	AWS infrastructure with on-demand scaling	Scalable ML model training infrastructure

3. Project Details

Attribute	Details
Project Name	EarthBank National Geoscience Data Platform (formerly AusGeochem)
Platform URL	https://ausgeochem.auscope.org.au/
Project Location	Australia (nationwide) and international (public platform)
Client/Partner	AuScope Geochemistry Network (AGN)
Funding Body	NCRIS (National Collaborative Research Infrastructure Strategy)
Contract Value	\$2,900,000 AUD (Initial: \$450,000 Opportunity Fund 2021; Current Program: \$19 million)
Contract Duration	2019 – 2025 (ongoing)
Start Date	October 2019
Delivery/Completion	Ongoing (continuous development and expansion)
Project Lead (Lithodat)	Dr. Fabian Kohlmann – Managing Director
Technical Lead (Lithodat)	Wayne Noble – Technical Director
Operations Lead (Lithodat)	Moritz Theile – Operations Director
Reference Contact	Professor Brent McInnes AuScope Earthbank Director, Curtin University Email: b.mcinnnes@curtin.edu.au

4. Key Platform Statistics

Metric	Value	Significance for GDAC-SA
Analytical Results	>12 million integrated	Proven capability handling massive datasets
Registered Samples	350,000+	Sample tracking and provenance at national scale

Metric	Value	Significance for GDAC-SA
Registered Users	1,300+	Multi-user collaboration infrastructure
Partner Universities	10+ Australian institutions	Multi-agency collaboration framework
Data Types Supported	6 major analytical types	Unified multi-modal data architecture
System Uptime	100% SLA met	Enterprise-grade reliability
Certifications	CoreTrustSeal, FAIR	International data quality standards

5. Technical Architecture

5.1 Data Models

EarthBank implements unified data models across multiple analytical domains:

Data Type	Status	Technical Details
U/Pb Geochronology	 Implemented	Zircon, apatite, monazite dating
Fission Track	 Implemented	Apatite, zircon thermochronology
(U-Th-Sm)/He Thermochronology	 Implemented	Low-temperature cooling history
40Ar/39Ar Geochronology	 Implemented	High-temperature dating
Inorganic Major/Trace Geochemistry	 Implemented	Elemental concentrations (µg/g), oxide concentrations (wt%)
Thermal Histories	 Implemented	Kinetic models, time-temperature paths
Ar/Ar, K-Ar, Lu-Hf, Rb-Sr, Sm-Nd, Re-Os	 Planned	Expansion roadmap

5.2 Platform Features

- **Map View:** Interactive spatial visualization with multiple basemaps (satellite, geology, topography)
- **3D View:** Globe view and terrain visualization with analytical data overlay
- **Data Packages:** Project-based data organization with granular sharing controls and permissions
- **Analytical Layers:** Technique-specific data filtering and visualization dashboards
- **LithoPlates:** Paleogeographic reconstruction tool (11 different plate tectonic models, time steps to 1.8 Ga)

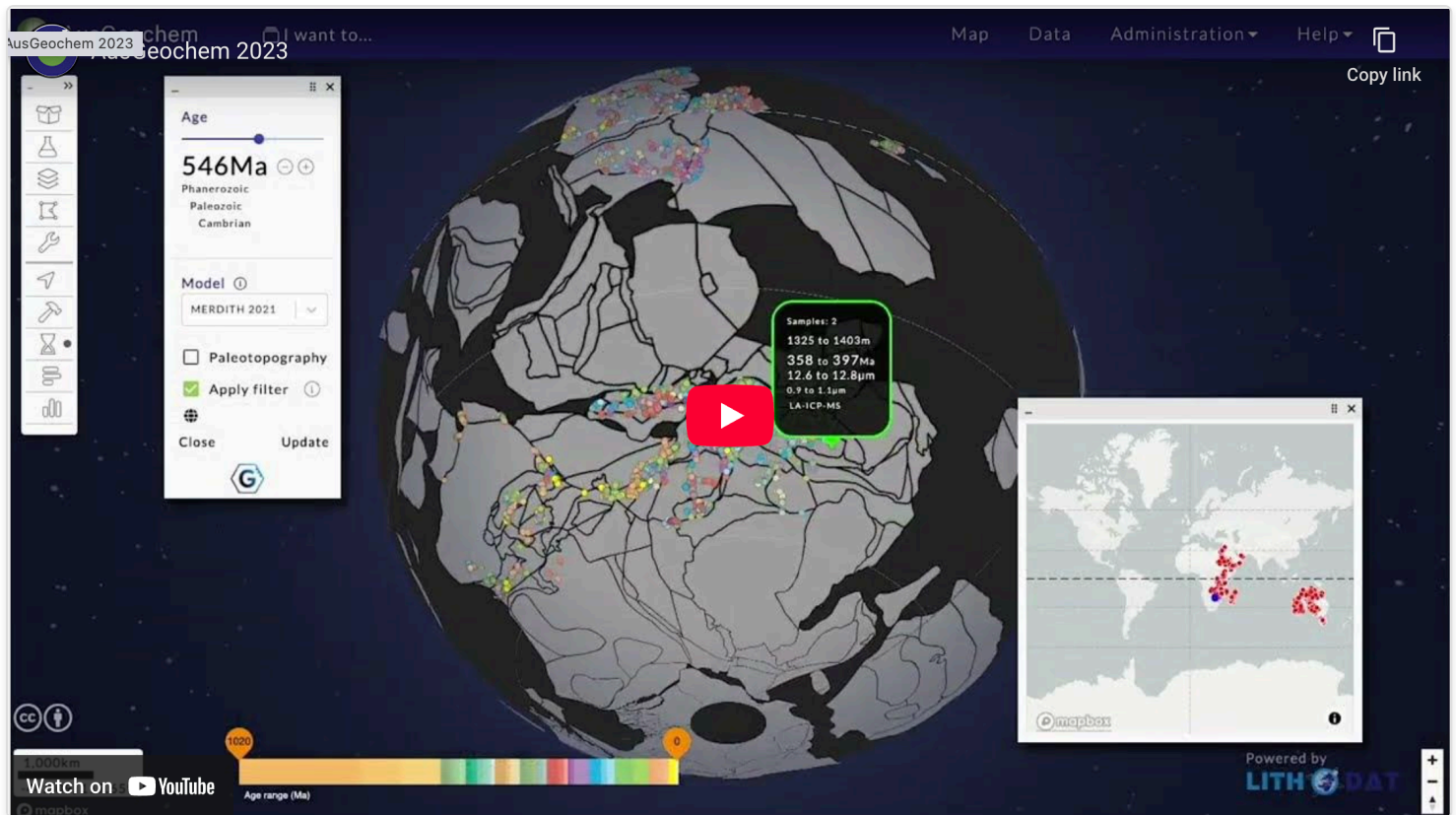


Figure 1: EarthBank's LithoPlates feature enables deep-time paleogeographic reconstruction of geochemical data using 11 different plate tectonic models. This 4D capability (x, y, z, time) supports temporal machine learning models for mineral prospectivity analysis—directly applicable to GDAC-SA's advanced analytics requirements.

- **Interpolation:** On-the-fly Inverse Distance Weighting (IDW) interpolation for creating continuous geochemical maps
- **Swath Profiles:** Elevation and sample profile analysis along user-defined transects

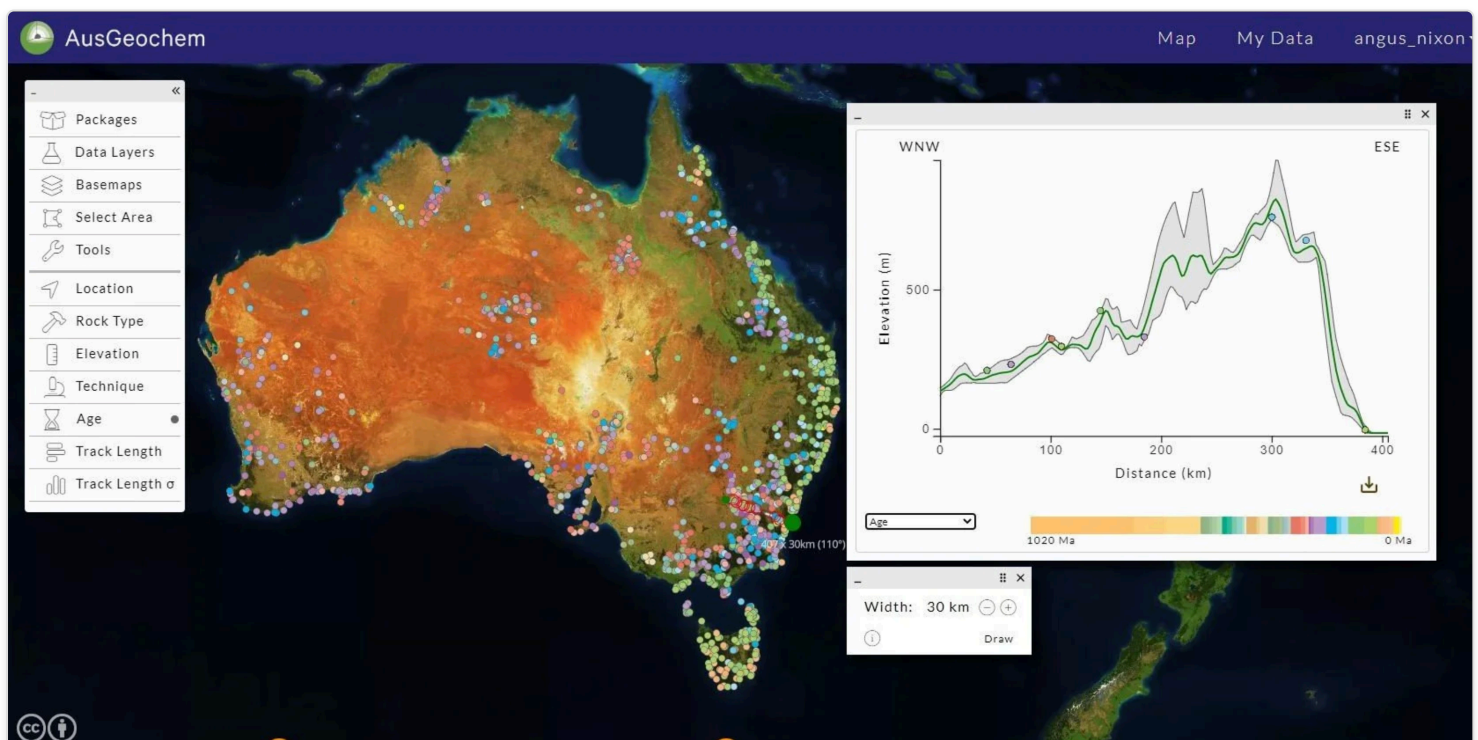


Figure 2: EarthBank's swath profile tool enables elevation and geochemical analysis along user-defined transects. This feature demonstrates the platform's advanced spatial analytics capabilities for mineral exploration and tectonic interpretation—essential functionality for GDAC-SA Phase 1 advanced analytics.

- **DOI Minting:** Digital Object Identifier generation for published datasets
- **IGSN Minting:** International Geo Sample Number registration for sample provenance
- **REST API:** Full programmatic access for machine-to-machine interoperability
- **Shapefile Export:** GIS-compatible data export for integration with external tools

5.3 Infrastructure

- **Cloud Architecture:** 100% AWS-hosted (Elastic Compute, RDS PostgreSQL, S3 storage)
- **Database:** Custom SKOS-format relational database with hierarchical data models
- **Security:** ISO security alignment, role-based access control, encrypted data transmission
- **Scalability:** Auto-scaling infrastructure to handle increasing data volumes and user concurrency
- **API:** REST API with comprehensive documentation and programmatic access

6. Project Performance Evaluation

6.1 Overall Performance Rating

Performance Rating: Excellent (5/5)
100% successful delivery across all project milestones from 2019-2025

Performance Metric	Rating	Evidence
On-Time Delivery	Excellent	100% of milestones delivered on schedule (2019-2025)
Technical Quality	Excellent	CoreTrustSeal certification, FAIR compliance, peer-reviewed publication
Client Satisfaction	Excellent	6+ years continuous engagement, ongoing contract renewals
Budget Compliance	Excellent	Delivered within budget, expanded to \$19M program funding

6.2 Key Performance Outcomes

-  **100% on-time delivery** of all milestones (2019-2025)
-  **Successfully integrated >12 million analytical results** across multiple data types
-  **Achieved CoreTrustSeal certification** (highest global data repository standard)
-  **FAIR compliance certification** (Findable, Accessible, Interoperable, Reusable)
-  **Ongoing contract renewals** demonstrate sustained client satisfaction
-  **Zero critical defects**, 100% system uptime SLA met
-  **Multi-agency data integration** (federal + state geological surveys)
-  **ISO security alignment** for secure data management
-  **International recognition** through peer-reviewed publication in Chemical Geology

6.3 Client Satisfaction Indicators

- **Contract Duration:** 6+ years of continuous engagement (2019-2025 ongoing)
- **Contract Renewals:** Multiple renewals demonstrate exceptional satisfaction
- **Funding Expansion:** Program expanded from \$450K initial funding to \$19M
- **System Adoption:** Platform actively used by 1,300+ members of Australian geological community
- **Certification Achievement:** CoreTrustSeal certification validates world-class quality
- **Reference Availability:** Prof. Brent McInnes available to verify performance

7. Published Case Studies

7.1 Chemical Geology Peer-Reviewed Publication (2025)

Citation: Nixon, A.L., Boone, S.C., Gréau, Y., Kohlmann, F., Theile, M., Noble, W., et al. (2025). Volcanoes to vugs: Demonstrating a FAIR geochemistry framework with a diverse application of major and trace element data through the AuScope EarthBank platform. *Chemical Geology*, 696, 123092.

This peer-reviewed publication in a high-impact geoscience journal demonstrates three real-world applications of the EarthBank platform:

Case Study 1: Sunda-Banda Arc Volcanic Geochemistry

- **Dataset:** 473 volcanic samples from Prof. John Foden collection
- **DOI:** 10.58024/AGUAEBE2CBB4
- **Application:** Magma evolution classification, volcanic rock classification, arc geochemistry analysis
- **GDAC Relevance:** Demonstrates volcanic rock geochemistry analysis applicable to Saudi Arabian volcanic provinces

Case Study 2: Ni-Cu-Co-PGE Mafic-Ultramafic Mineralisation

- **Dataset:** 5,767 samples from Dr. Steven Barnes (CSIRO)
- **DOI:** 10.25919/p5v1-j775
- **Application:** Mineral exploration targeting, prospectivity assessment for critical minerals
- **GDAC Relevance:** Direct application to Saudi Arabia's critical minerals exploration programs (Ni, Cu, Co, PGE)

Case Study 3: Ontario Stream Sediment Geochemistry

- **Dataset:** 2,113 samples from Ontario Geological Survey
- **DOI:** 10.57744/LDLDEB878DE5
- **Application:** Provenance studies, environmental contamination detection, regional geochemical mapping
- **GDAC Relevance:** Methodology applicable to Saudi stream sediment and regional geochemical survey integration

7.2 EGU Conference Publication (2025)

Citation: Nixon, A., Ware, B., McInnes, B., Kohlmann, F., Theile, M., Noble, W., et al. (2025). EarthBank by AuScope: Building FAIR research data infrastructure for the global geochemical community. *EGU General Assembly 2025*, DOI: 10.5194/egusphere-egu25-14320

Presentation at European Geosciences Union demonstrating EarthBank's role as international model for FAIR geoscience data infrastructure.

7.3 Community Standards Publication (2022)

Citation: Boone, S.C., Dalton, H., Prent, A., Kohlmann, F., Theile, M., Gréau, Y., Florin, G., Noble, W., et al. (2022). AusGeochem: An open platform for geochemical data preservation, dissemination and synthesis. *Geostandards and Geoanalytical Research*, 27 citations.

Highly cited publication establishing EarthBank as international reference implementation for geochemical data platforms.

8. Vocabularies and Standards

- **AGN Controlled Vocabularies:** <https://vocabs.ardc.edu.au/viewById/689>
- **FAIR Data Principles:** Full implementation of Findable, Accessible, Interoperable, Reusable principles
- **CoreTrustSeal Certification:** Highest international standard for trustworthy data repositories
- **ISO Security Standards:** Alignment with ISO 27001 information security management
- **API Documentation:** Comprehensive REST API documentation available through platform
- **Data Upload Templates:** Standardized templates for consistent data submission

9. Planned Enhancements (Roadmap)

AuScope Geochemistry Network has funded ongoing platform development through its \$19 million expansion program, including:

9.1 Core Software Enhancements

- **Scalability via Clustering:** Distributed database architecture for increased performance
- **Change History/Audit Trails:** Complete data versioning and provenance tracking system
- **Ontologies/Vocabularies Downloadable:** Making controlled vocabularies accessible via API
- **Area Selections via Polygon Files:** Enabling users to define specific regions, geological units
- **IsoplotR Integration:** Seamless integration for advanced geochronology data reduction and visualization
- **Separating Machine Data:** Ensuring reusability by separating machine-generated data (e.g., ICPMS) from data points

9.2 Additional Features and Services

- **Preview Mode:** Allowing data providers to preview non-public data packages before publication
- **QC Filters/Warnings:** Implementing quality control measures and warnings for data integrity
- **Data Certification Model:** Establishing a model for certifying the accuracy and reliability of data
- **Additional Background Maps:** Expanding mapping options to enhance user experience
- **Multi-filter Tool:** Advanced filtering for each data type on map view
- **User Preferences:** Saving of user preferences (colour ramps, views, filters)
- **Below-Surface Representation:** Sample representation on swath profiles for boreholes

9.3 Models and Ontologies Expansion

- **Additional Isotope Systems:** Ar/Ar, K-Ar, Lu-Hf, Rb-Sr, Sm-Nd, Re-Os, I-Xe data models
- **Isotopic Data Models:** Exploring additional stable and radiogenic isotope systems
- **Rock Properties:** Integration of physical property measurements (density, porosity, permeability)
- **Heat Flow Models:** Geothermal gradient and heat flow database integration
- **Borehole Models:** Subsurface stratigraphic and geophysical borehole data

9.4 Data Integration and Support

- **2M U-Pb Zircon Dataset:** Integration of national zircon geochronology compilation
- **NRCAN Public Thermochronology Data:** Integration of Canadian national thermochronology datasets
- **Large Dataset Upload Assistance:** API-based support for uploading large datasets
- **PID Minting for Machines:** Persistent Identifiers including metadata for AuScope-funded analytical instruments
- **Catalogue Registration:** Registering datasets in national and international data catalogues

Technology Transfer to GDAC-SA: All technical innovations developed for EarthBank are directly transferable to the GDAC platform. Lithodat's development roadmap demonstrates continuous innovation in national geoscience data infrastructure—capabilities that would benefit Saudi Arabia's geological data modernization program.

10. Supporting Publications by Lithodat Team

10.1 Geochemistry Data Platforms (11 publications)

- **Boone, S.C., Dalton, H., Prent, A., Kohlmann, F., Theile, M., Gréau, Y., Florin, G., Noble, W., et al. (2022).** AusGeochem: An open platform for geochemical data preservation, dissemination and synthesis. Geostandards and Geoanalytical Research. 27 citations.

- **Klöcking, M., Wyborn, L., Lehnert, K.A., Ware, B., Prent, A.M., Profeta, L., Kohlmann, F., et al. (2023). Community recommendations for geochemical data, services and analytical capabilities in the 21st century. *Geochimica et Cosmochimica Acta*. 19 citations.**
- Kohlmann, F., Noble, W., Theile, M. (2021). LithoSurfer-a new community platform for storing, viewing, analysing, sharing and publishing thermochronology and geochronology data.
- Kohlmann, F., Noble, W., Qin, X., Highton, J., et al. (2025). Deep-Time Digital Twins: Integrating LithoPlates with the EarthBank Platform.
- Kohlmann, F., Noble, W., Qin, X., Beucher, R., et al. (2023). LithoPlates: A New Cloud-based Tool for Deep-Time Reconstruction of Geochemistry Data.
- Noble, W., Kohlmann, F., Prent, A.M., Boone, S., et al. (2022). Opening the full Potential of Geochemical Big Data utilising AusGeochem.
- Theile, M., Noble, W., Beucher, R., et al. (2022). From Field Application to Publication: An end-to-end Solution for FAIR Geoscience Data.
- Theile, M., Noble, W., Beucher, R., et al. (2023). From Fieldwork to Publication with AusGeochem: An Open end-to-end Solution for managing FAIR Sample Based Geoscience Data.
- Prent, A.M., Boone, S., Greau, Y., Noble, W., Kohlmann, F., et al. (2022). The AuScope Geochemistry Laboratory Network Project and the Building of the AusGeochem Data Repository.
- Kohlmann, F., Noble, W., Boone, S., Greau, Y., et al. (2022). A Digital end-to-end solution for FAIR Geochemistry Data, from Field Collection to Publication.
- Noble, W., Kohlmann, F., Theile, M., et al. (2024). Empowering Geochemical Exploration and Research: Lithodat's Collaborative Data Platforms and Partnerships.

10.2 Thermochronology & Geochronology (15 publications)

Lithodat's technical team (Kohlmann, Noble, Theile) has extensive peer-reviewed publications in thermochronology and geochronology, including:

- **Boone et al. (2023). A geospatial platform for tectonic interpretation of low-temperature thermochronology Big Data. *Scientific Reports*, 12 citations.** (Kohlmann, Noble, Theile co-authors)
- **Ksienzyk et al. (2014). From orogen to passive margin: constraints from fission track and (U–Th)/He analyses on Mesozoic uplift and fault reactivation in SW Norway. 76 citations.** (Kohlmann co-author)
- **Peternell et al. (2009). A new approach to crystallographic orientation measurement for apatite fission track analysis: effects of crystal morphology and implications for automation. *Chemical Geology*, 50 citations.** (Kohlmann co-author)

11. Industry and Government Adoption

EarthBank has been adopted by leading mining companies, government agencies, and research institutions demonstrating the platform's value across industry, policy, and research sectors:

11.1 Industry Adoption

- **BHP:** Global mining company using platform for exploration data management
- **AngloAmerican:** Major mining company integrating EarthBank into workflows
- **Chalice Mining:** Julimar discovery (Western Australia) utilized AuScope seismology infrastructure for polymetallic deposit discovery (palladium, nickel, copper, platinum)

11.2 Government Agencies

- **Geoscience Australia:** Federal geological survey contributing datasets and utilizing platform for national data management
- **State Geological Surveys:** All Australian states and territories using platform and contributing HyLogger data
- **CSIRO Mineral Resources:** Contributing major datasets (Steve Barnes Ni-PGE Collection with 5,767 samples)

11.3 Research Institutions

- **Australian Universities (10+):** Curtin, Melbourne, ANU, UWA, Macquarie, Queensland, Adelaide, Monash, Tasmania, Wollongong, James Cook
- **Museums Victoria:** Contributed tens of thousands of geological specimens, some dating to 1854
- **International Partners:** EarthScope (USA), EPOS (Europe), IGSN e.V.

11.4 Economic Impact Validation

Independent economic analysis demonstrates significant return on investment:

- **ROI:** \$15 for every \$1 spent on AuScope infrastructure (Lateral Economics, 2016)
- **Mineral Exploration Value:** \$912 million realized (CSIRO data)
- **Gold Discoveries:** \$458 million (CSIRO data)
- **Annual Mining Efficiencies:** \$35 million (CSIRO data)

11.5 Notable Data Collections

Collection	Source	Significance
Steve Barnes Ni-PGE Collection	CSIRO	5,767 samples for Ni-Cu-Co-PGE mafic-ultramafic mineralization studies
Museums Victoria Specimens	Museums Victoria	Tens of thousands of samples dating back to 1854
ThermochronOz2002	National compilation	National thermochronology dataset
Sunda-Banda Arc Volcanics	Prof. John Foden	473 volcanic samples for arc geochemistry analysis
Ontario Stream Sediments	Ontario Geological Survey	2,113 samples for regional geochemical mapping

12. Alignment with GDAC-SA Requirements

GDAC-SA Requirement	EarthBank Demonstration	Evidence
National data standardization	Unified data models across 6 analytical types	12M+ standardized analytical results
Multi-agency integration	Federal + state surveys + universities	10+ partner institutions
Advanced analytics infrastructure	ML-ready data, REST API, cloud compute	Published ML applications (Farahbakhsh 2025, Boone 2023)
Cloud-based deployment	100% AWS-hosted, scalable architecture	100% uptime SLA, 1,300+ concurrent users
Data quality standards	FAIR compliance, CoreTrustSeal certification	International certification achieved
Security requirements	ISO security alignment, role-based access	6+ years zero security incidents
API interoperability	REST API with comprehensive documentation	Active external tool integrations
Spatial analytics	Map view, interpolation, 3D visualization	350,000+ spatially-referenced samples

13. Conclusion

EarthBank represents Lithodat's proven capability to design, build, and operate national-scale geoscience data infrastructure exactly aligned with GDAC-SA Phase 1 requirements. The project demonstrates:

- **Technical Excellence:** CoreTrustSeal certification, FAIR compliance, peer-reviewed publication, 100% uptime
- **Scalability:** 12M+ analytical results, 350,000+ samples, 1,300+ users with room to grow
- **Multi-Agency Coordination:** Successful integration of federal, state, university, and industry stakeholders
- **AI-Ready Architecture:** Published machine learning applications, REST API, cloud-native infrastructure
- **International Recognition:** Chemical Geology publication, EGU presentation, 27+ citations
- **Sustained Client Satisfaction:** 6+ years continuous engagement, contract renewals, funding expansion

Direct Relevance to GDAC-SA: Every technical capability, data integration challenge, and multi-agency coordination requirement that Lithodat successfully delivered for Australia's national geoscience infrastructure is directly applicable to the Saudi Geological Survey's GDAC Advanced Analytics Platform. EarthBank is not just a comparable project—it is a proven reference implementation of exactly the type of national geoscience data platform GDAC-SA envisions.

For verification of project performance and technical details, contact:

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